

FRONTIER APPLICATIONS

From a surface that remembers to a habitat that repairs itself



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Framework attribution

18 expanded architectures for analog, digital, physical and future systems.

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Companion to Perpetuum Mobile PM 1.0. Prepared with Codex assistance.

Cover: AI-generated concept art showing a proposed adaptive workshop. It is not a photograph, engineering drawing or evidence of a built PM machine.

01 / PURPOSE

From an idea list to an application book

This companion documents and expands the preceding 18-application reply. The original wording is preserved in a printed appendix and an embedded Markdown witness. Each expanded application now has a use scenario, operating mechanism, model, source mapping, demonstrator, comparison criteria and future extension.

A01-A18 are application identifiers. V01-V07 refer to the existing families in **Perpetuum Mobile PM 1.0**. The new numbering does not imply eighteen new physical laws or replace the earlier specification.

The central proposition

A common formal description could connect digital rules, physical memory, geometric transformations, measured feedback and repair. Its value must appear in the useful result: a better experiment, a more reliable action, lower resource use or a clearer explanation of failure.

How to read the expanded designs

Proposal: a new integration described here. **Component evidence:** a cited result supporting one ingredient. **Model:** a conditional equation requiring parameters and an operating region. **Demonstrator:** a finite experiment that has not been performed for this book. **Future:** an extension dependent on unresolved engineering.

Equations are compact design contracts. They do not replace a full material model, hardware specification, control proof or qualification process. Numeric acceptance targets are proposed experiment criteria unless a cited source explicitly reports a measured result.

The underlying PM 1.0 PDF is embedded unchanged. It contains the four original source PDFs, the exact SEED contract and scoped reference checks. The new book introduces no claim that these eighteen complete systems have been built.

02 / READING MAP

Application atlas / 1-9

Application	Embodiment / PM family	Pages
A01 Habitats that inspect and repair themselves	digital, analog, physical V02, V05, V06, V07	7-8
A02 Robot skin that computes through its material	analog, physical, digital V02, V03, V07	9-10
A03 A workshop table that becomes its own tooling	digital, physical, analog V01, V03, V05	11-12
A04 A programmable optical nervous system	analog, digital, physical V04, V02, V07	13-14
A05 Remote manufacturing from a "construction seed"	digital, physical, analog V01, V05, V07	15-16
A06 VR environments with physical, moving geometry	digital, physical, analog V01, V03, V07	17-18
A07 Rooms with programmable acoustics	analog, digital, physical V04, V07	19-20
A08 Laboratories that reconfigure their own experiments	digital, physical, analog measurement V01, V05, V07	21-22
A09 Printers that grow structures under continuous inspection	digital, physical, analog sensing V05, V01, V07	23-24

Applications occupy pp. 7-42. Integration and experiment planning follow on pp. 43-48; the original reply is on pp. 49-51. The bibliography and provenance record follow. Internal PDF bookmarks provide direct chapter navigation.

02 / READING MAP

Application atlas / 10-18

Application	Embodiment / PM family	Pages
A10 Persistent worlds synchronized through rules and edits	digital, networked V01, V07	25-26
A11 Deployable structures that retune themselves	physical, digital, analog sensing V03, V06, V07	27-28
A12 Repair swarms that tolerate communication gaps	physical, digital, networked V07, V05, V06	29-30
A13 Infrastructure with distributed physical memory	Analog sensing, Physical memory, Digital coordination V02, V07	31-32
A14 Greenhouses that coordinate survival and maintenance	Physical cultivation, Digital prediction, Resource scheduling V06, V07	33-34
A15 Clothing with a changing mechanical response	Mechanical material, Pneumatic actuation, Digital interaction V03, V01	35-36
A16 Planetary equipment that manufactures its own fixtures	Physical fabrication, Robotic assembly, Digital geometry V05, V07, V01, V06	37-38
A17 Machines that test their own design alternatives	Digital experiment design, Physical testing, Feedback fabrication V01, V05, V07	39-40
A18 Combined wave-and-material computers	Photonic propagation, Material memory, Digital calibration V04, V02	41-42

Applications occupy pp. 7-42. Integration and experiment planning follow on pp. 43-48; the original reply is on pp. 49-51. The bibliography and provenance record follow. Internal PDF bookmarks provide direct chapter navigation.

03 / SHARED OPERATING MODEL

What the common architecture contributes

Source/family	Concrete contribution	Additional work in an application
UGTS / geometry and lineage	Typed objects, ordered hinges, support, compatibility, guarded events	Device/material bindings and valid event certificates
atomOS / logic and records	Packed support, explicit JK/ASA/NA profiles, transactional digital state	Actuator decoding and any newly introduced feedback policy
WANTWOMBAN / material feedback	Word-command-heat-colour-measurement loop	Calibrated device response, energy boundary and physical state
PM SEED / digital replay	Exact input generation and reproducible logical streams	Full descriptor, external-input transcript and numerical rules

```
descriptor + state + input -> proposal -> guard -> commit
actual command -> physical evolution -> measured readback
```

The physical plant keeps evolving while software evaluates a proposal. Digital commit cannot reverse deposited material, heat, charge or motion. A command and its acknowledgment must therefore be separate records, followed by reconciliation with the actual state.

The full causal loop includes sensor, computation, communication and actuator delay. A future machine needs a stability or bounded-error argument for the resulting system. Reusing the same record format does not prove that two physical components are compatible.

Source map: PM 1.0 pp. 4-14 and 34-36. Literal OTAN2, code-mixing OTAN2, geometric atan2, numerical H4 and Boolean XOR retain their separate meanings.

04 / RESEARCH PRIORITIES

Three flagship directions

1 / A surface with physical memory

Combine A02 material memory with A03 adaptive tooling. Begin by showing that a measured history-dependent signal improves a bounded contact task. Add deformation only after separating the influence of temperature, shape, load and readout drift. This is a compact place to test whether the source-specific composition adds value.

2 / A workshop that changes its capabilities

Combine the adaptive table, A08 experiment automation and A09 inspected fabrication. A fixture is designed, made, measured and used; its later removal or failure remains in the physical record. The result would be a workshop that configures itself for particular tasks within known machine and material limits.

3 / A habitat that commissions repairs

Combine A01, A12 and A14: resource monitoring, intermittent-link robots and local maintenance. Begin with an uncrewed installation and a small approved repair catalog. The final ambition is a system that closes inspection, diagnosis, planning, repair and verification while accounting for consumables.

These priorities are editorial judgments about useful research structure, not measured rankings of readiness or commercial value. A01 is the broadest integration; A02 and A03 offer smaller experiments that can expose its assumptions.

A future photonic or acoustic layer can be added through A04/A18 after its bandwidth, calibration, power and conversion costs are known. Its contribution should be evaluated against the same task performed electronically.

APPLICATION / DESIGN

A01 / Habitat inspection and repair

Habitats that inspect and repair themselves
digital + analog + physical / V02, V05, V06, V07

A scene to make the idea concrete

An unoccupied remote habitat notices that a service panel is warming differently after each equipment cycle. The proposed system sends an inspection robot, compares measurements with the component history, and selects replacement or local repair. After the work, independent observations determine whether service can resume. Every action consumes tracked energy, material, time and remaining component life.



Operating mechanism

The new architecture links telemetry, material-memory features and a geometric inspection map to a bounded repair planner. NASA ISAAC demonstrated robotic inspection triggered by synthetic fault telemetry; its fuller manipulator response was not performed in that ISS scenario [A01R1]. Vision-controlled jetting establishes a separate precedent for measuring deposited geometry and correcting subsequent material delivery [A01R2]. PM would connect these capabilities through typed component identities, joint-test requirements and resource budgets. Issued commands enter the record as pending physical events. Acknowledgments and inspection establish what occurred; their absence cannot erase physical effects or justify assuming success.

```

E_next = E + E_in - E_service - E_repair - E_waste
repair_ok = geometry_ok AND joint_ok AND budget_ok
history_next = append(history, observed_work)
  
```

E denotes electrical storage; energy terms are joules. Outgoing terms partition expenditures without counting losses twice. Acceptance flags summarize specified inspections and operating limits, with missing evidence treated as unknown. This is a supervisory contract, not a strength law. A rejected repair retains its material, heat and mechanical consequences.

Where this framework enters

PM pages 18-19 supply the proposed temporal material readout; pages 24-25 distinguish planned deposition from material already present. Pages 26-29 connect resource accounting with distributed observations. The UGTS registry supplies component and joint definitions, atomOS supplies admission records, and WANTWOMBAN supplies the powered measurement loop. Page 8 requires recovery from measured physical state after an interrupted command.

APPLICATION / VALIDATION

A01 / Experiment and future

Habitats that inspect and repair themselves

Minimum demonstrator

Use a non-crewed enclosure containing a replaceable fan duct, a monitored thermal load and a sacrificial repair coupon. Inject repeatable obstructions and sensor faults, then compare fixed maintenance scheduling with inspection-triggered action. Require an independent camera or gauge to verify the result. Add material only to the coupon until separate bond tests justify expansion. Interrupt one acknowledgment deliberately and evaluate whether the system discovers what physically happened before continuing.

Measure	Proposed comparison or criterion
Fault localization	On held-out faults, report detection delay and location error against independent inspection, including missed faults and false dispatches at matched sampling effort.
Repair validity	Accept recovery only when predefined geometric and joint tests pass; report every failed repair, rework action and unresolved outcome across repeated trials.
Resource value	Compare downtime, consumed material and total energy with conventional maintenance under identical faults, counting inspection travel, computation, cleanup and unsuccessful repairs.

The failure that matters

A warm patch can indicate changed airflow, sensor drift or an unrelated load. A visually complete deposit can still have a weak bond. The planner therefore separates diagnosis, action and verification. If their assumptions fail, it preserves the observed damage and repair history and suspends the affected service.

How the future extension would differ

A larger installation could share repair knowledge across habitats while retaining each site's actual calibration and wear. Modular service corridors could make inspection and replacement easier than repairing inaccessible structures. The transformative possibility is coordinated upkeep without continuous supervision; the decisive unresolved work is reliable physical diagnosis, joint qualification and resource-aware recovery over long missions.

The exact role of SEED

SEED reproduces synthetic fault schedules, candidate repair geometries and planning comparisons from a pinned manifest. It does not recreate the habitat's damage, wear or sensor readings; those remain measured inputs with timestamps and uncertainty.

Component evidence: [A01R1], [A01R2]. See the bibliography for the result supported and the boundary of that evidence.

APPLICATION / DESIGN

A02 / Material-computing robot skin

Robot skin that computes through its material

analog + physical + digital / V02, V03, V07

A scene to make the idea concrete

A gripper touches an unfamiliar object. Instead of reducing each contact to an instantaneous pressure value, the proposed skin retains a fading trace of pressure and temperature changes. That physical history helps a trained readout distinguish a steady hold from repeated rubbing or emerging slip, while a separate controller decides whether and how to adjust the grip.



Operating mechanism

Pressure sensors would drive a nearby material-memory layer through calibrated electrical or optical signals. Tan and colleagues demonstrated pressure encoding and integration using an optoelectronic spiking architecture and photomemristors [A02R1]. Du and colleagues demonstrated dynamic-memristor temporal processing with a software readout [A02R2]. The proposed skin combines these component ideas with replaceable sensing patches and explicit preparation procedures. Each patch emits normalized features, uncertainty and age; a digital readout estimates contact class. Local actuation consumes that estimate only through a separately validated policy, rather than treating a classifier score as mechanical authority.

```

tau_i * zdot_i = -z_i + F_i(p,T,h)
I_i = Read(z_i,p,T; calibration)
y_hat = W_out * [1; Normalize(I)]
  
```

Here z is a dimensionless fitted memory state, p pressure in pascals, T temperature in kelvin, and I measured current. Positive τ sets a fitted response time. This relaxation model is a local surrogate; material history h and the functions F and Read require device-specific characterization.

Where this framework enters

PM pages 18-19 define material memory, readout training and comparisons against memoryless sensing. Pages 20-21 separate surface kinematics from force and actuator laws. UGTS contributes patch identities, compatibility and ordered geometry; atomOS records admitted commands and old/new logical state. WANTWOMBAN contributes calibrated readback and retained material history. Pages 28-29 address stale information when patches share observations.

APPLICATION / VALIDATION

A02 / Experiment and future

Robot skin that computes through its material

Minimum demonstrator

Mount four replaceable patches on a stationary gripper fixture and record synchronized pressure, current and independent contact video. Apply steady presses, ordered taps and controlled lateral motion across held-out objects. Train the readout before evaluating those objects. Compare material features with raw instantaneous sensing and a digital temporal baseline. Repeat after bending the carrier and changing ambient conditions within the characterized range, while retaining identical task definitions and recording preparation time.

Measure	Proposed comparison or criterion
Temporal discrimination	Measure held-out contact-class error against independent video labels, comparing memoryless and digital temporal baselines with identical sensors, tasks and data splits.
Operating drift	Report calibration change and prediction error after repeated bending and thermal cycles; predefined limits determine when a patch must undergo recalibration.
System burden	Count sensing, encoding, material stimulation, readout, training and preparation energy together; require any claimed advantage to survive the full matched-task comparison.

The failure that matters

Material memory can retain irrelevant contacts, drift with temperature or change after deformation. Read pulses may disturb the state being measured. A failed patch must become unavailable rather than silently clear. Grip adjustments remain bounded by independent force and position observations, and a digital reset does not reset the material automatically.

How the future extension would differ

A distributed computing skin could mix fast electrical memory with slower thermal channels, allowing different patches to specialize in contact history and environmental change. Patches might exchange compact features while keeping raw evidence for uncertain events. The remaining challenge is durable, calibratable computation under real deformation, with a useful overall benefit after electronics and maintenance are included.

The exact role of SEED

SEED fixes contact sequences, learning splits and digital probe masks for comparisons. Physical initialization still requires a measured preparation procedure; equal seeds cannot guarantee equal conductance, temperature, deformation or contact history across patches.

Component evidence: [A02R1], [A02R2]. See the bibliography for the result supported and the boundary of that evidence.

APPLICATION / DESIGN

A03 / A table that becomes tooling

A workshop table that becomes its own tooling

digital + physical + analog / V01, V03, V05

A scene to make the idea concrete

An operator places an irregular component on a flat workbench. The proposed table scans it, raises selected supports and forms a cradle for inspection or light assembly. Where the actuated surface cannot supply the needed feature, a dispenser adds a removable fixture. The table measures contact and loading before presenting the arrangement as usable tooling.



Operating mechanism

A height-actuated array with compliant contact pads provides a changing support surface. MIT's inFORM demonstrated shape-based affordances, constraints and object actuation [A03R1]; it does not establish machining-grade fixtures. PM would add calibrated load sensing, a constrained support planner and optional fixture deposition. The planner first solves for attainable heights, then tests contact under the intended load. Orthogonal Hadamard probe patterns can identify small local response changes, but cannot establish nonlinear strength. A deposited fixture becomes part of the obstruction and cleanup model even when its support test fails.

```

z_star = argmin_z ||A*z - h_target||^2
subject to z_min <= z <= z_max
accept_support = geometry_ok AND all(f_i <= f_limit_i)
E_in = integral(sum_i V_i * I_i) dt
  
```

Pin heights z and target heights h are metres; A is a calibrated dimensionless interpolation. After placement, measured forces f in newtons must pass limits for the intended load. Voltage and current determine electrical energy. Feasible geometry alone does not certify stiffness, friction or adhesion.

Where this framework enters

PM pages 16-17 provide reproducible target geometry and persistent edit identity. Pages 20-21 connect ordered UGTS hinges with calibrated actuation, loading and probe patterns. Pages 24-25 describe guarded deposition and retained physical history. atomOS supplies proposal and disposition records, while WANTWOMBAN supplies the measured feedback pattern. Page 13 requires physical conformance beyond a successful geometric calculation.

APPLICATION / VALIDATION

A03 / Experiment and future

A workshop table that becomes its own tooling

Minimum demonstrator

Build a small replaceable pin or tile array for metrology and light assembly, with force sensing and an independent pose camera. Test a fixed library of irregular coupons, including shapes absent from calibration. Compare adjustable manual supports with automatically generated cradles. Add one removable fixture only after measuring its material response. Track setup time, pose drift and cleanup, and deliberately fail one support test to check that the obstruction remains recorded.

Measure	Proposed comparison or criterion
Fixture accuracy	Measure component pose and drift under predefined light loads, using independent metrology and held-out shapes; report every geometry accepted outside tolerance.
Setup effort	Compare complete setup and removal time with conventional adjustable supports, including scanning, calibration, failed attempts, deposited material curing and required cleanup.
Useful reuse	Count successful reconfigurations before recalibration or replacement, and report energy, consumables and surface damage per accepted fixture across the same workload.

The failure that matters

Sparse supports can concentrate stress, and a good height match can hide slipping or weak attachment. Friction, compliance and sensor offsets change with use. The table must withhold acceptance when measured support differs from its model, preserving actual contact geometry and deposited material while returning to a tested recovery state.

How the future extension would differ

The same architecture could grow into configurable inspection benches, adaptive assembly stations and molds for carefully qualified processes. A robot might request a fixture by function and receive a measured capability record instead of assuming a universal work surface. The hardest extension is supporting larger forces while preserving reliable contact models and economical maintenance between tasks.

The exact role of SEED

SEED recreates the target fixture family, probe sequence and planning comparison. Measured component shape, pin calibration, wear and deposited fixtures remain external state; the seed specifies a requested support, not its demonstrated load capacity.

Component evidence: [A03R1]. See the bibliography for the result supported and the boundary of that evidence.

APPLICATION / DESIGN

A04 / Programmable optical signal network

A programmable optical nervous system

analog + digital + physical / V04, V02, V07

A scene to make the idea concrete

A sensing instrument needs different signal combinations as its task changes. The proposed optical network physically routes coherent inputs, mixes their amplitudes and sends selected outputs to detectors. When a path drifts, the controller measures the change and searches for another configuration. Its changing connections resemble a nervous system functionally, without implying biological cognition or autonomous understanding.



Operating mechanism

Tunable interferometers form the physical mixing network, while electronic drivers set phase and coupling. Programmable silicon meshes have demonstrated multiple optical functions [A04R1], and later work demonstrated feedback-based characterization, routing and configuration [A04R2]. PM would represent every usable channel, calibration and failed element in one typed graph. A bounded compiler fits a requested forward transfer matrix to measured device behavior. Coherent detectors verify amplitude and phase; power-only readout cannot certify signed arithmetic. Commands remain proposals until the settled device has been measured, and reconfiguration consumes recorded electrical energy.

```

b(omega) = S_fwd(omega;theta,T)*a(omega)
S_target = diag(phi,1/phi,1,-1)*H4/phi
S_fwd^dagger*S_fwd <= I
  
```

Amplitudes a and b are power-normalized, in square-root watts, across four inputs and four outputs. Tuning is held fixed during measurement. The golden ratio ϕ scales the pinion to a contractive target. Reflection, reverse transmission and loss ports remain separate parts of the physical account.

Where this framework enters

PM pages 22-23 define the wave architecture and the passive realization constraint for WANTWOMBAN's numerical pinion. H4 supplies a concrete orthogonal mixing target; atomOS Boolean XOR is a different operation. UGTS contributes compatible ports and ordered maps. Pages 28-29 add distributed observation age and command history. The proposed compiler must preserve these boundaries rather than treating graph edges as fabricated waveguides.

APPLICATION / VALIDATION

A04 / Experiment and future

A programmable optical nervous system

Minimum demonstrator

Calibrate a four-input, four-output optical bench or mesh, including phase references, loss and tuner settling. Implement H4 and the scaled pinion, then probe basis vectors and previously unseen coherent superpositions. Introduce a bounded tuning offset or disable one path. Compare recalibration and alternative routing against a fixed configuration. Count detector, laser, control and tuning energy, and retain unsuccessful searches instead of reporting only configurations that converged.

Measure	Proposed comparison or criterion
Transform fidelity	Measure complex matrix error on held-out coherent inputs over the declared bandwidth, retaining phase uncertainty, insertion loss and all rejected configurations.
Reconfiguration recovery	After a predefined path disturbance, report time and measured success rate for restoring the target within tolerance, including searches that never converge.
Complete cost	Compare useful task output with a digital baseline while counting lasers, detectors, calibration, tuning and conversion; claim benefit only within tested workloads.

The failure that matters

Phase drift, thermal crosstalk or a missing reference can invalidate a previously good transform. A passive network cannot supply the unscaled pinion's maximum amplitude gain. The controller must report unavailable channels or a failed fit, and must not relabel attenuation compensation in software as physical amplification.

How the future extension would differ

A larger network could link sensor arrays to specialized material-memory stages, routing signals according to measured task needs and device health. Distributed instruments might exchange compact configuration descriptions while keeping local calibration authoritative. The unresolved questions are stable inverse tuning, scalable monitoring and whether the complete optical-electronic system improves a useful objective under realistic drift and conversion costs.

The exact role of SEED

SEED reproduces probe vectors, optimizer initialization and digital routing choices. Optical phase, temperature, fabrication offsets and detector noise remain measured state. Repeating the seed alone does not recreate the same physical transfer matrix.

Component evidence: [A04R1], [A04R2]. See the bibliography for the result supported and the boundary of that evidence.

APPLICATION / DESIGN

A05 / Manufacturing from a construction seed

Remote manufacturing from a "construction seed"

digital + physical + analog / V01, V05, V07

A scene to make the idea concrete

A remote workshop receives a compact construction package for a replacement support. It resolves the exact rulebook, checks available material and measures the installation site. The proposed system regenerates a design, adapts only permitted dimensions, manufactures the support and returns evidence of its actual geometry and tested interfaces. The package coordinates construction; it does not supply matter.



Operating mechanism

A construction seed selects a design inside a versioned bounded grammar; the manifest binds geometry, materials, joints and acceptance procedures. NASA ARMADAS provides ground evidence of autonomous assembly from discrete structural blocks [A05R1]. Vision-controlled jetting provides a separate measured-deposition approach [A05R2]. PM would make these alternative manufacturing adapters consume the same typed intent while retaining different physical laws. Each accepted operation references measured predecessor geometry and available resources. A failed joint or interrupted command remains in the site history, and local measurements determine whether to continue, rework or request another design.

```

plan = Generate(seed, manifest, measured_site)
accept_j = geometry_j AND joint_test_j AND resource_j
mass_next = mass_prev + mass_added - mass_removed
  
```

The manifest fixes the generator and permitted adaptations; measured site data are explicit inputs. Acceptance flags carry their evidence and unknown status. Mass terms are kilograms of the tracked part, with added retained material distinguished from feedstock waste. Identical plans do not establish identical physical parts.

Where this framework enters

PM pages 11-12 define exact SEED framing, manifests and replay. Pages 16-17 preserve generated object identity and edits. Pages 24-25 require actual deposition and support status to remain distinct; pages 28-29 address remote command acknowledgment. UGTS supplies grammar and relation records, atomOS supplies admission bookkeeping, and WANTWOMBAN contributes the explicit command-to-measurement boundary rather than an automatic fabrication law.

APPLICATION / VALIDATION

A05 / Experiment and future

Remote manufacturing from a "construction seed"

Minimum demonstrator

Use two workstations and one instrumented assembly or printing cell. Send a seed package for several small support variants, including a declared fit adjustment based on local scanning. Regenerate the nominal plan independently before building. Test dimensions and joints with independent gauges or witness coupons. Interrupt a transfer and an acknowledgment, then check restart behavior. Compare package size, setup effort and accepted-part cost with ordinary model-file delivery under identical production constraints.

Measure	Proposed comparison or criterion
Digital agreement	Require independent regeneration to match canonical geometry and operation identities from the same complete manifest, with the first divergence recorded when inputs differ.
Physical acceptance	Evaluate held-out parts against declared dimensional and joint tests, reporting rejects and rework separately; a matching design hash cannot replace these measurements.
Delivery value	Compare total transmitted bytes, setup time and accepted-part resources with ordinary model delivery, including rulebook distribution, calibration, local adaptations and failed builds.

The failure that matters

An unavailable generator version, substituted material or stale site scan can invalidate the package. A completed command with a lost acknowledgment can otherwise be repeated physically. The receiver therefore resolves content before planning, assigns operation identities, and reconciles actual work before retrying or accepting an adapted continuation.

How the future extension would differ

Qualified construction packages could support remote maintenance fleets, modular infrastructure and local manufacture of specialized tools. A shared grammar might reduce repeated design transmission when many variants use the same rules. Expansion requires interoperable calibration, reliable joint evidence and controlled adaptation; an arbitrary object cannot be assumed recoverable from a small seed without its defining information.

The exact role of SEED

The seed chooses reproducible variants and operation sequences within a pinned design family. Material supply, machine calibration, site measurements and physical deviations belong in separate records. Source identity and authorship metadata do not substitute for those records.

Component evidence: [A05R1], [A05R2]. See the bibliography for the result supported and the boundary of that evidence.

APPLICATION / DESIGN

A06 / VR with moving physical geometry

VR environments with physical, moving geometry

digital + physical + analog / V01, V03, V07

A scene to make the idea concrete

A virtual workshop contains a shelf, a railing and a shaped workpiece. The proposed room moves tracked physical proxies into selected contact locations so the user can touch real surfaces. As the virtual layout changes, the system schedules reachable rearrangements and verifies their actual poses. Visual continuity never becomes permission to assume that physical geometry has teleported.



Operating mechanism

Mobile or height-actuated proxies provide a limited physical vocabulary that a virtual scene can reuse. RoomShift demonstrated furniture-moving robots for room-scale haptics [A06R1], while HapticBots explored multiple moving, shape-changing contact proxies [A06R2]. PM would add persistent virtual identities, bounded physical mapping and explicit contact-readiness records. A planner predicts upcoming interactions, reserves movement corridors and issues commands only within the characterized tracking and stopping envelope. The renderer obtains readiness from measured pose and settled motion, not from the command queue. Unreachable contacts receive a different interaction rather than fabricated physical support.

```

d_clear >= d_stop + e_tracking + v_rel*latency
proxy_target = Map(virtual_contact, calibrated_room)
contact_ready = pose_ok AND settled AND tracked
  
```

Distances are metres; v_{rel} bounds relative closing speed throughout latency in seconds. d_{stop} bounds remaining relative closing distance during stopping, including continued user motion. Tracking error and stopping behavior require measured bounds. This simplified corridor condition needs geometry-specific verification; it cannot alone guarantee collision avoidance.

Where this framework enters

PM pages 16-17 supply reproducible virtual geometry, stable identities and edits. Pages 20-21 distinguish an ordered hinge model from the mechanics of moving furniture or proxies. Pages 28-29 preserve observation age and interrupted-command history. UGTS supplies compatible contact mappings; atomOS records proposals and readiness. WANTWOMBAN contributes calibrated feedback, while page 8 prevents digital rollback from erasing real motion.

APPLICATION / VALIDATION

A06 / Experiment and future

VR environments with physical, moving geometry

Minimum demonstrator

Begin with a seated, bounded interaction area and two small tracked proxies, using an independent tracker and measured stopping behavior. Reproduce a fixed virtual task with and without physical rearrangement. Test contact alignment before introducing users, then evaluate only characterized motions and loads. Inject delayed tracking and an interrupted command to examine readiness handling. Log virtual targets, actual trajectories, contacts and unavailable interactions, including time spent waiting for physical movement.

Measure	Proposed comparison or criterion
Contact alignment	Measure physical-to-virtual contact error and readiness delay over held-out layouts, using independent tracking and reporting contacts withheld because alignment was unavailable.
Motion handling	Across predefined latency and interruption trials, verify declared stopping and readiness behavior from measured trajectories, counting every violated bound and uncertain outcome.
Interaction value	Compare task performance and reported experience with fixed props under identical virtual tasks, including waiting, unavailable contacts, setup effort and actuator noise.

The failure that matters

Tracking loss or delayed motion can put a proxy somewhere other than its virtual counterpart. Furniture can retain momentum after a software pause. The interaction must become unavailable when readiness assumptions fail, and recovery must use the measured room state rather than replaying old movement commands or restoring virtual coordinates.

How the future extension would differ

Larger spaces could reuse configurable walls, work surfaces and lightweight props for design review, training or remote collaboration. Seeded scenes could share a virtual plan while each room solves its own physical realization. The hardest boundary is consistent interaction across different geometries, motion limits and tracking conditions without concealing unsupported contact or imposing excessive rearrangement delays.

The exact role of SEED

SEED recreates virtual layouts and proposed interaction sequences. Each room retains its own dimensions, proxy inventory, calibration and measured motion. A shared seed therefore synchronizes digital intent, while contact readiness remains a local physical decision.

Component evidence: [A06R1], [A06R2]. See the bibliography for the result supported and the boundary of that evidence.

APPLICATION / DESIGN

A07 / Programmable acoustic rooms

Rooms with programmable acoustics

analog + digital + physical / V04, V07

A scene to make the idea concrete

Two teams share a workshop, but each needs a different audio channel. A proposed room adjusts resonant wall panels to strengthen intended speaker-to-listener paths and reduce interference elsewhere. Microphones measure the result as furniture and occupants move. The aim is bounded acoustic zoning over selected frequencies, with explicit retuning intervals and electrical costs.



Operating mechanism

Motorized resonator cavities change boundary impedance, altering the room's measured acoustic transfer matrix. Research has demonstrated acoustic channel isolation using tunable resonators installed at room boundaries [A07R1]. The proposed system adds a versioned target map, microphone calibration and a bounded optimizer that selects panel configurations. It tests small candidate changes, accepts measured improvements and retains a known configuration when observations become unreliable. Speakers supply acoustic energy; motors and computation consume additional power. Passive boundaries redistribute and absorb sound; they cannot guarantee silence everywhere or arbitrary broadband channel separation.

```

p(w) = H(w,theta)*a(w)
theta_next = argmin_theta J(H_measured,H_target)
J = error_band + lambda*switching_cost
  
```

Here p and a are calibrated pressure amplitudes in pascals, making H dimensionless under the selected convention. Frequency w and panel configuration θ are explicit. The objective uses normalized error and switching cost. The model assumes approximately linear acoustics and stationary boundaries during each measurement.

Where this framework enters

PM pp. 22-23 provide V04 transfer-matrix fitting and the distinction between digital routing and actual wave coupling. UGTS assigns typed acoustic ports, ordered configurations and permitted connections. atomOS masks admit selected panel commands; WANTWOMBAN supplies the command/readback pattern through an acoustic observation adapter. PM pp. 28-29 add timestamped microphone observations and stale-data handling across distributed controllers.

APPLICATION / VALIDATION

A07 / Experiment and future

Rooms with programmable acoustics

Minimum demonstrator

Build a small enclosure with two speakers, four microphones and several independently adjustable resonators. Characterize the empty enclosure, then repeat with movable absorbing blocks. Optimize only a declared narrow frequency band and fixed listener locations. Compare static panels, conventional speaker equalization and the proposed boundary controller. Reserve previously unseen configurations for evaluation. Meter all speaker, motor and controller energy. Proposed criteria below are experiment targets, not existing performance claims.

Measure	Proposed comparison or criterion
Channel isolation	Improve median desired-to-unwanted mean-square-pressure ratio by at least 3 dB over static panels across the predefined held-out positions and frequency band.
Retuning stability	Complete each accepted reconfiguration within the declared settling budget, with no measured improvement reported from samples acquired before panel motion has settled.
System value	Report total energy and added latency alongside acoustic gain; retain adaptive boundaries only when the predefined benefit outweighs their measured operating costs.

The failure that matters

A person entering a previously calibrated zone can invalidate the transfer matrix. Resonators also have bandwidth and travel limits, and microphones may confuse source changes with room changes. Failed validity checks must suspend optimization, retain bounded commands and record the unresolved acoustic condition.

How the future extension would differ

A later building could share room models across partitions, allowing meeting areas and listening zones to change with occupancy. Combining acoustic panels with movable furniture could broaden available configurations. The research gap is dependable adaptation over changing listeners, multiple sound sources and wide bandwidths without excessive measurement disturbance, motor noise or maintenance burden.

The exact role of SEED

SEED fixes probe order, candidate configuration sampling and held-out evaluation scenes. Actual microphone signals, moving occupants and motor positions remain external records. Replaying the same seed reproduces the experiment design, not the room's physical sound field.

Component evidence: [A07R1]. See the bibliography for the result supported and the boundary of that evidence.

APPLICATION / DESIGN

A08 / Laboratories that reconfigure experiments

Laboratories that reconfigure their own experiments

digital + physical + analog measurement / V01, V05, V07

A scene to make the idea concrete

A materials laboratory repeatedly measures how a harmless test coating responds to temperature and illumination. Instead of repeating one fixed recipe, a proposed robotic workcell chooses its next measurement, moves samples between calibrated stations and changes permitted instrument settings. Each change carries a reason, resource budget and traceable specimen identity, enabling reproducible experimental campaigns.



Operating mechanism

A mobile manipulator or gantry transports specimens between dispensing, heating and optical measurement stations. A published mobile robotic chemist demonstrates autonomous use of laboratory workstations within a defined workflow [A08R1]. Here, reconfiguration means selecting validated routes, fixtures and settings from a finite capability library. An experimental-design policy ranks candidate measurements by predicted information gain and cost. Before execution, device compatibility, calibration validity and specimen availability are checked. Actual commands and observations update the model; an unexpected result can redirect the next experiment without rewriting what already occurred.

```

x_next = argmax_x [IG(x|D)-lambda*Cost(x)]
subject to x in ValidCapabilities(state)
y_next = Measure(Execute(x_next))
D <- D union {(x_next,y_next)}
  
```

Candidate x describes a complete permitted experiment, including station settings and specimen identity. D contains observed data and uncertainties. Information gain IG and cost are normalized under a declared objective. The policy proposes experiments; it does not certify scientific truth or authorize capabilities absent from the registry.

Where this framework enters

PM pp. 5-8 supply typed contracts, snapshot/proposal/commit and physical acknowledgment. UGTS turns instrument capabilities and workflow dependencies into inspectable definitions. WANTWOMBAN supplies the calibrated temperature/material/readback cycle. atomOS records admitted operations and observations without equating logical bits with measurements. V05 on PM pp. 24-25 contributes irreversible specimen history; V07 on pp. 28-29 handles device messages and interrupted acknowledgments.

APPLICATION / VALIDATION

A08 / Experiment and future

Laboratories that reconfigure their own experiments

Minimum demonstrator

Use a gantry, temperature-controlled stage, calibrated light source and camera to measure benign reference patches. Predefine fixture choices and exposure/temperature ranges. Compare a fixed grid with adaptive selection under identical measurement and energy budgets. Introduce a deliberately stale calibration and one interrupted station acknowledgment. Evaluate predictions against independent held-out measurements. Reproduce the decision sequence from recorded observations, while maintaining separate records for specimens consumed, altered or rejected.

Measure	Proposed comparison or criterion
Information efficiency	Reach the preregistered held-out prediction error using at least 20 percent fewer accepted measurements than the fixed-grid campaign under identical operating limits.
Trace completeness	Every accepted observation must resolve to a specimen, calibration, actual device setting and command acknowledgment; missing provenance prevents promotion into validated campaign data.
Recovery behavior	After injected communication loss, reconcile actual station and specimen state before continuing; no completed physical operation may be silently duplicated during tested recoveries.

The failure that matters

An optimizer can exploit measurement artifacts instead of learning the coating response. Instrument drift, sample aging and fixture changes can masquerade as discoveries. The campaign therefore separates model uncertainty from apparatus faults and stops proposing dependent measurements when calibration or specimen history becomes unresolved.

How the future extension would differ

A future laboratory could assemble larger experimental workflows from interoperable instrument contracts and compare competing physical models during the same campaign. Robots might exchange qualified modules between stations, extending beyond changing settings. The unresolved challenge is validating those new configurations and maintaining scientific controls as hardware, samples and hypotheses change together over long campaigns.

The exact role of SEED

SEED fixes candidate sampling, initial design points and tie-breaking. Measured outcomes enter the transcript and can change later selections. A seed without observations, instrument versions and sample preparation records cannot reconstruct either the adaptive decision path or the experiment.

Component evidence: [A08R1]. See the bibliography for the result supported and the boundary of that evidence.

APPLICATION / DESIGN

A09 / Printing under continuous inspection

Printers that grow structures under continuous inspection

digital + physical + analog sensing / V05, V01, V07

A scene to make the idea concrete

A printer builds a branching scaffold whose upper segments depend on the quality of earlier joints. The proposed machine scans newly deposited material and adjusts subsequent flow or path placement. It extends another branch only after its attachment region satisfies declared geometry and support conditions, keeping rejected material visible in the model and fabrication history.



Operating mechanism

A dispenser deposits a selected curable material while local scanning measures height and footprint. Vision-controlled jetting has demonstrated correction of subsequent deposition using measured surface geometry [A09R1]. This proposal adds a finite branch grammar and explicit support admission. The controller estimates deposition yield and shrinkage, compares measured geometry with the target, and modifies bounded commands for the next segment. Cure and adhesion require separately calibrated evidence; a correct visible outline is insufficient. Inspection is repeated at a declared sampling rate, with unresolved intervals recorded rather than described as uninterrupted observation.

```

h_next = h + eta*Q*dt/A - shrinkage
alpha_dot = k(T,I)*(1-alpha), k >= 0
extend = geometry_ok AND support_ok AND clearance_ok
  
```

Height h and shrinkage use metres, Q is volume flow, and A is footprint area. Yield η is fitted. The illustrative cure fraction α requires formulation-specific kinetics. The Boolean extension decision combines measured or certified conditions, with uncertainty margins; it cannot infer strength from shape alone.

Where this framework enters

PM pp. 24-25 define V05 deposition, inspection and accepted-support states. UGTS provides parametric sweeps, candidate attachment neighborhoods and durable branch lineage. atomOS binds intended masks to actual fabrication records. WANTWOMBAN supplies causal material feedback through a replacement height/process readout. V01 on PM pp. 16-17 supplies the finite target grammar; V07 supports separate scanner and dispenser acknowledgments.

APPLICATION / VALIDATION

A09 / Experiment and future

Printers that grow structures under continuous inspection

Minimum demonstrator

Print two-layer lattices containing straight segments and Y junctions with one calibrated material. Use independent microscopy and witness specimens to establish acceptable geometry and support. Introduce repeatable flow disturbances within the selected envelope. Compare fixed deposition, ordinary height feedback and the proposed guarded branch controller. Measure retained mass and waste. Record any occluded region, incomplete scan or failed support decision before allowing the next dependent segment to proceed.

Measure	Proposed comparison or criterion
Geometry correction	Reduce median absolute height error by at least 25 percent against the fixed-path baseline, while adding no more than 10 percent fabrication time.
Support admission	Accept no attachment that fails the predefined independent witness check in the demonstration set; report trial counts and unresolved support decisions explicitly.
Material accounting	Keep retained-mass balance residual within propagated measurement uncertainty, including waste and rejected deposits; compare additional scanning energy against the achieved reduction in rework.

The failure that matters

A geometrically correct joint may contain an uncured layer, weak interface or hidden void. Sensors can also miss a defect behind the nozzle. The machine must preserve rejected deposits as obstructions and avoid extending unsupported branches while independent inspection or a validated process model remains unavailable.

How the future extension would differ

A later system could inspect an existing part, generate a repair branch around its measured defect and fabricate only the missing material. Multiple deposition and scanning robots could share the same specimen history. The critical gap is reliable online evidence of bonding and load capacity, especially when materials, access angles and thermal histories vary.

The exact role of SEED

SEED generates target scaffolds, disturbance schedules and comparison cases. The actual deposited geometry, curing history and material consumption remain measured inputs. Regenerating an identical planned scaffold does not reproduce an identical physical structure without matching fabrication conditions.

Component evidence: [A09R1]. See the bibliography for the result supported and the boundary of that evidence.

APPLICATION / DESIGN

A10 / Persistent worlds through shared rules

Persistent worlds synchronized through rules and edits

digital + networked / V01, V07

A scene to make the idea concrete

Several people revisit the same virtual workshop from different devices. Most objects are regenerated from shared rules, while their placements, annotations and modifications persist as recorded edits. The proposed world exchanges semantic changes instead of repeatedly transmitting complete geometry, preserving durable object identities and showing unresolved conflicts when two users alter the same dependent structure.



Operating mechanism

A finite generator expands a pinned grammar and seed into canonical object records. Ordered edit layers modify those records; clients render them through their own adapters. OpenUSD already provides composable scene-description layers with explicit composition semantics [A10R1]. The proposed synchronization layer adds command identities, rule hashes, dependency invalidation and an authoritative ordering policy for conflicting edits. Independent changes may merge only under declared commutation rules. A changed generator creates a versioned migration problem, not permission to silently reinterpret old objects. Canonical world state is compared before device-specific rendering.

```

base = Generate(seed,descriptor)
world = Replay(base,CanonicalOrder(accepted_edits))
replica_ok = Hash(Canonical(world_A)) == Hash(Canonical(world_B))
  
```

The descriptor pins algorithms, assets, numeric rules and finite budgets. Accepted edits include actor, object, operation and causal identity. Hash equality checks serialized state integrity under the chosen encoding; it does not establish that conflict resolution matches user intent or that different renderers produce identical pixels.

Where this framework enters

PM pp. 11-12 specify exact SEED framing and replay; pp. 16-17 define V01 editable generated worlds. UGTS provides definition DAGs and durable lineage. atomOS supplies canonical occupancy groups and storage adapters. WANTWOMBAN contributes optional pinion and seam motifs. K1 evidence remains bounded implementation evidence on PM p. 32; V07 supplies versioned messages and stale-state reconciliation on pp. 28-29.

APPLICATION / VALIDATION

A10 / Experiment and future

Persistent worlds synchronized through rules and edits

Minimum demonstrator

Run two independently implemented clients against a bounded scene containing branching objects, one portal and editable annotations. Disconnect one client, make overlapping and independent edits, then reconnect. Compare canonical objects, lineage and accepted edit order before rendering. Repeat with a deliberately changed grammar hash. Measure descriptor, edits, cache and transmitted bytes against a fully materialized scene baseline with identical persistence semantics. Include deletion, reparenting and dependent-geometry invalidation cases.

Measure	Proposed comparison or criterion
State convergence	Require identical canonical integer object records after replaying the same accepted transcript on both implementations, including deletion, dependency changes and reconnect scenarios.
Conflict fidelity	Present every unresolved noncommuting edit conflict with its original operations intact; no tested reconnection may silently discard a user's accepted semantic change.
Storage benefit	Count descriptor, asset, edit, transcript and cache bytes together; claim compression only when total storage beats the materialized baseline at matched semantics.

The failure that matters

Identical seeds cannot repair missing assets, incompatible arithmetic or divergent grammar versions. Automatic ordering can also produce a technically valid but unwanted edit result. Clients must reject unresolved descriptors and distinguish explicit conflict decisions from ordinary replay, preserving the original operations for later review.

How the future extension would differ

The same architecture could connect persistent training environments, design spaces and digital twins across headsets and workstations. Physical observations could become additional versioned events without overwriting the generative model. The main research gap is durable semantic migration: old edits must retain understandable meaning when object rules, simulation models and collaborating software evolve over years.

The exact role of SEED

SEED defines the reproducible base world and named procedural streams. Human edits remain explicit events, including their conflict decisions. Neither a seed nor a hash compresses arbitrary unrecorded creative changes into recoverable information.

Component evidence: [A10R1]. See the bibliography for the result supported and the boundary of that evidence.

APPLICATION / DESIGN

A11 / Deployable structures that retune

Deployable structures that retune themselves

physical + digital + analog sensing / V03, V06, V07

A scene to make the idea concrete

A shelter or instrument support unfolds, measures its loaded shape, and makes small adjustments as wind or temperature changes. The structure separates deployment movements from alignment corrections. Sensors report hinge positions, strain and thermal state, while the controller retains a bounded operating envelope and records of locks, contacts and power.



Operating mechanism

Foldable members connect through joints and mechanical latches. Motors or thermally activated elements perform deployment; reversible trim actuators handle later corrections. Research combining shape-memory polymers with bistability demonstrates programmed deployment and shape reconfiguration [A11R1]. It does not establish unrestricted repeated retuning. This proposal uses measured joint and strain feedback to select bounded corrections after deployment, updating its stiffness/contact model when a latch changes state. A support path must remain load-bearing throughout each transition. Thermal actuation requires heating and cooling accounts, while stored spring energy and external loads remain explicit.

```

M(q)*qddot + C(q,qdot)*qdot + D*qdot
+ grad_q U(q,latch) = tau + F_ext
q_target = argmin_q [shape_error + lambda*actuation_cost]
commit_latch only if load_path_ok AND measured_lock_ok
  
```

Generalized coordinates q describe joints; U is elastic and gravitational potential for the current latch configuration. M is inertia; $C*qdot$ contains velocity-dependent inertial terms, and D damping. τ and F_{ext} are actuator and other generalized loads. Travel, stress and stability constrain both targets and transitions.

Where this framework enters

PM pp. 20-21 connect UGTS ordered hinges to separately calibrated mechanics, including rooted per-panel transforms and passive coupling through failed cells. WANTWOMBAN supplies local thermal feedback when thermal actuators are chosen. atomOS records command admission and latch state. V06 on PM pp. 26-27 accounts for finite mission resources; V07 retains distributed sensor age and acknowledgment history.

APPLICATION / VALIDATION

A11 / Experiment and future

Deployable structures that retune themselves

Minimum demonstrator

Build a small four-panel frame with instrumented hinges, reversible trim motors and independently monitored latches. Apply known dead loads and controlled disturbances after deployment. Compare fixed geometry with feedback trimming under equal power limits. Track panel positions using an external camera and strain using independent sensors. Include a failed latch indication and a sensor delay. Repeatedly cycle the structure within a predefined envelope, recording fatigue indicators and any irreversible deformation.

Measure	Proposed comparison or criterion
Shape regulation	Keep held-out loaded-shape error below the preregistered tolerance while respecting measured joint travel and stress limits throughout every accepted correction in the demonstration.
Transition integrity	Do not accept a latch change without independent lock confirmation and a validated load path; record all rejected transitions and observed structural excursions.
Resource advantage	Compare useful alignment improvement with total deployment, trimming, sensing and cooling energy; require the declared benefit at equal load capacity and mission duration.

The failure that matters

A local position reading can conceal a failed lock or redistributed load elsewhere. Temperature can change stiffness and actuator response simultaneously. The controller must stop dependent movements when its load-path model is invalid, retaining passive structural support rather than assuming a digital latch flag carries force.

How the future extension would differ

Larger systems could become temporary shelters, adjustable antenna supports or deployable instrument platforms that compensate slow distortion. Distributed nodes could coordinate repair inspections with their geometry records. The unresolved engineering work is qualification of transitions, fatigue, environmental exposure and fail-passive load support, particularly when thermal and mechanical aging change the calibrated model during a mission.

The exact role of SEED

SEED generates repeatable deployment sequences, disturbance cases and optimization starting points in simulation. Actual latch engagement, strain and aging require observations. A repeated command sequence cannot guarantee repeated geometry after material or contact properties have changed.

Component evidence: [A11R1]. See the bibliography for the result supported and the boundary of that evidence.

APPLICATION / DESIGN

A12 / Repair swarms across communication gaps

Repair swarms that tolerate communication gaps

physical + digital + networked / V07, V05, V06

A scene to make the idea concrete

Three robots inspect and patch an isolated installation where radio links repeatedly disappear behind structures. The proposed team shares a repair map when connected, but each robot retains bounded local authority and verified specimen history. A disconnected robot can finish an already reserved task or inspect another region, while uncertain shared work waits for reconciliation.



Operating mechanism

Each agent stores a local map, resource state, task reservation and actual-command ledger. Independent sensors maintain local clearance and verify completed work. JPL's NeBula describes multi-robot autonomy with uncertainty-aware navigation and mesh networking that accommodates intermittent links [A12R1]. This proposal extends that component basis with exclusive repair-region reservations, bounded offline actions and post-reconnection evidence merging. Peers exchange versioned observations instead of assuming a plan was executed. A lost acknowledgment triggers inspection of actual repair state. Material deposition remains irreversible, so duplicate commands need physical reconciliation as well as message deduplication.

```

uncertainty(age) = e0 + v0_bound*age + 0.5*a_bound*age^2
offline_ok = reserved AND local_clearance AND resources_ok
resume_shared = reconcile(commands,measurements,repair_state)
  
```

The positional envelope assumes a bounded initial position error, initial speed and subsequent acceleration. Reservation ownership must remain valid under a specified lease or persistent assignment protocol. Local clearance uses current sensing. The rules permit bounded offline work; they do not imply agreement during arbitrary partitions.

Where this framework enters

PM pp. 28-29 provide V07 command identities, stale-data envelopes and separate digital/physical replay. UGTS supplies local support, compatibility and repair lineage. atomOS records admitted task and actuator states. WANTWOMBAN supplies measured actuation feedback. V05 on PM pp. 24-25 preserves deposited/rejected material; V06 on pp. 26-27 limits battery, consumables and mission duration independently of successful communication.

APPLICATION / VALIDATION

A12 / Experiment and future

Repair swarms that tolerate communication gaps

Minimum demonstrator

Use three tabletop mobile agents around an instrumented mock installation, initially placing removable markers instead of structural patches. Assign separate repair regions and track all robots independently. Inject bounded delay, duplicate messages, disconnections and a lost completion acknowledgment. Compare centralized stop-on-disconnect behavior with the proposed local policy under identical task and battery budgets. After reconciliation succeeds, repeat selected cases with benign deposition and inspect whether any physical task was inadvertently duplicated.

Measure	Proposed comparison or criterion
Task completion	Complete more predefined independent tasks than the stop-on-disconnect baseline during the same outage schedule, without exceeding equal energy, material or local-motion limits.
Duplicate prevention	Execute no repair twice in the fault trial set; verify physical outcomes independently and report unresolved cases rather than counting acknowledgments as completion.
Evidence convergence	After connectivity returns, agents must resolve completed work to compatible specimen histories and reservation ownership before accepting new actions in overlapping repair regions.

The failure that matters

Unbounded partitions can prevent exclusive ownership from being safely reassigned. Stale geometry, clock uncertainty or hidden partial repairs may invalidate an otherwise reasonable local plan. The system therefore narrows authority during disconnection and preserves uncertain work as unresolved until fresh sensing and an explicit ownership protocol permit continuation.

How the future extension would differ

A larger swarm could maintain remote power installations, inspect deployed structures and fabricate small repairs using shared generative maps. Some robots might relay messages while others inspect or carry material. The central gap is coordinating physical exclusivity and trustworthy repair evidence under changing contacts, failed agents and prolonged communication loss without depending on a permanently reachable coordinator.

The exact role of SEED

SEED fixes simulated faults, initial task allocation and deterministic tie-breaking over stable identifiers. Real message arrival times and partial physical actions remain logged events. Replaying digital decisions requires that transcript; matching the repaired installation additionally requires independent physical verification.

Component evidence: [A12R1]. See the bibliography for the result supported and the boundary of that evidence.

APPLICATION / DESIGN

A13 / Infrastructure that retains recent loads

Infrastructure with distributed physical memory

Analog sensing + Physical memory + Digital coordination / V02, V07

A scene to make the idea concrete

A bridge carries replaceable sensing modules that retain short histories of local strain and temperature. During a communication outage, each module continues recognizing unusual sequences and preserves selected event records. After reconnection, engineers receive a spatially organized account of recent loading, sensor uncertainty and maintenance needs. This proposed memory layer supplements conventional structural assessment.



Operating mechanism

Distributed fibre sensing and digital twins have already monitored real railway bridges [A13R1]. Separately, dynamic memristors have demonstrated temporal information processing [A13R2]. The proposed integration converts calibrated strain signals into bounded electrical stimuli for local memory cells. Their relaxation provides short temporal features; a digital logger retains durable event records. A trained readout distinguishes specified loading patterns, then neighboring modules exchange timestamped summaries. Temperature compensation, preparation history and drift remain explicit inputs. Neither a cell's retained state nor an anomaly score directly measures crack size or structural capacity.

```

dh_i/dt = -h_i/tau_i + k_i*abs(deps_i/dt)
y_i = W_i*[1; h_i; normalized_temperature_i]
tau_i > 0
  
```

This is a proposed local fading-memory surrogate, not a universal memristor law. Strain ϵ and memory h are dimensionless; τ is seconds and k is dimensionless. Measured device responses must justify its operating range. Durable damage estimates require a separate validated structural model.

Where this framework enters

PM pages 18-19 supply the V02 material-state/readout separation and comparisons against memoryless sensors. Pages 28-29 supply V07 message identity, stale-data limits and local operation. UGTS locates each module and its calibration lineage; atomOS records admitted samples and outcomes. The WANTWOMBAN feedback pattern supplies the observation cycle through a strain adapter, rather than treating its colour model as structural mechanics.

APPLICATION / VALIDATION

A13 / Experiment and future

Infrastructure with distributed physical memory

Minimum demonstrator

Instrument a laboratory beam at four positions using independent strain gauges and temperature references. Feed recorded loading sequences into replaceable memory modules while logging raw signals centrally. Hold out complete loading sessions for evaluation. Introduce communication interruptions and sensor offsets, then compare memoryless, digital recurrent and material-memory readouts under equal sampling budgets. Independent beam measurements determine whether reported events correspond to actual imposed changes, rather than sensor artefacts.

Measure	Proposed comparison or criterion
Held-out event recognition	Match a preregistered detection target and false-alarm limit on unseen loading sessions, including temperature shifts and repeated device preparations.
Outage record completeness	Recover every locally committed event identifier after the specified outage, with timestamps, uncertainty and explicitly recorded missing raw samples.
Total monitoring cost	Compare energy, communication volume and maintenance effort against conventional sensing at matched detection quality, including memory preparation and readout.

The failure that matters

Seasonal drift can resemble accumulated loading, while relaxed memory can erase a relevant event before transmission. A disconnected or saturated sensor may remain electrically plausible. Recovery therefore marks uncertain intervals and requests conventional inspection when the calibrated interpretation fails; local confidence never authorizes an unsupported structural capacity claim.

How the future extension would differ

A larger installation could maintain a map of component histories across bridges, tunnels or industrial frames, selecting inspections where independent evidence is weakest. The decisive research question is whether physical temporal features reduce communication and power enough to justify replacement, calibration and drift management. Field endurance and engineering interpretation remain separate qualification tasks.

The exact role of SEED

SEED fixes synthetic loading libraries, experiment ordering and evaluation splits. Real strain, temperature, communication interruptions and memory preparation remain recorded inputs. Replaying the same seed reconstructs the stimulus plan, not the bridge's past physical state.

Component evidence: [A13R1], [A13R2]. See the bibliography for the result supported and the boundary of that evidence.

APPLICATION / DESIGN

A14 / Greenhouses coordinating crops and repairs

Greenhouses that coordinate survival and maintenance

Physical cultivation + Digital prediction + Resource scheduling / V06, V07

A scene to make the idea concrete

A greenhouse coordinates crop conditions with the condition of its pumps, screens, lamps and water stores. When equipment degrades, the controller adjusts the cultivation schedule and reserves resources for repair. The proposed system keeps a finite crop mission inside declared operating limits while showing growers which biological, mechanical or resource assumptions have become uncertain.



Operating mechanism

Wageningen's AGROS trials use crop and climate models, sensor feedback and digital-twin decisions for cucumber cultivation; some crop actions still require greenhouse staff [A14R1]. The proposed extension adds equipment-state estimates and maintenance scheduling to that control loop. Root-zone water, air conditions and storage levels constrain candidate actions. A fault changes the available actuator set and the forecast, rather than merely raising an alert. Local controllers retain a characterized fallback during network loss. Crop response, disease and component degradation require separate fitted models and observations.

```

dW_root/dt = irrigation - uptake - evaporation - drainage
C_T*dT/dt = Q_light + Q_heat - Q_vent - Q_loss
P_lamps + P_pumps + P_service <= P_available
  
```

W_{root} is root-zone water mass, with flows in kilograms per second. Uptake enters plant storage; transpiration is a separate plant loss. C_T is effective thermal capacity and Q terms are watts for a declared boundary. The electrical budget is instantaneous. Crop stress and maintenance require additional models.

Where this framework enters

PM pages 26-27 provide V06 resource balances, mission horizons and degraded operating modes. Pages 28-29 provide V07 acknowledged commands and stale-observation rules. UGTS represents beds, pipes and replaceable devices with versioned connections; atomOS records competing schedules and committed actions. WANTWOMBAN contributes a measured actuation/readback pattern, while crop physiology and irrigation physics enter through independently calibrated domain adapters with their own units.

APPLICATION / VALIDATION

A14 / Experiment and future

Greenhouses that coordinate survival and maintenance

Minimum demonstrator

Use two small cultivation zones with independently measured irrigation, drainage, temperature and electrical consumption. Establish normal operating envelopes before introducing bounded disturbances. First emulate pump failure with a water-only bench; then test approved reduced-capacity schedules on a finite crop trial. Compare ordinary climate control with the proposed maintenance-aware scheduler. Record manual actions and crop observations explicitly, and assess both plant-condition outcomes and equipment recovery over the same declared horizon.

Measure	Proposed comparison or criterion
Environmental service continuity	Keep independently measured crop-zone variables within preregistered limits during the specified disturbance set, with all excursions reported by duration.
Resource balance closure	Reconcile water and electrical inputs, storage changes and outputs within declared measurement uncertainty throughout cultivation, fallback operation and maintenance.
Maintenance scheduling value	Compare recovery time, crop outcomes and resource use against ordinary control under matched faults, including human intervention and replacement costs.

The failure that matters

A biased moisture sensor can produce confident but harmful irrigation recommendations, while maintenance may consume capacity needed for cooling. Biological deterioration can also exceed a climate model's assumptions. The scheduler must expose uncertainty, preserve independent observations and invoke the declared grower intervention when crop or equipment models cease to apply.

How the future extension would differ

Cooperating greenhouses could share heat, water-treatment capacity and maintenance resources while retaining local crop authority. Their schedules would negotiate real resource ports and delivery uncertainty, not exchange abstract promises of surplus. The open challenge is demonstrating useful coordination across weather, crop stages, equipment faults and changing biological conditions over repeated seasons.

The exact role of SEED

SEED generates reproducible weather scenarios, simulated equipment faults and controller comparisons. Actual weather, crop development, maintenance completion and valve acknowledgments remain external evidence. Seeds support controlled comparison without claiming that living plants are reproducibly initialized digital objects.

Component evidence: [A14R1]. See the bibliography for the result supported and the boundary of that evidence.

APPLICATION / DESIGN

A15 / Clothing with adjustable resistance

Clothing with a changing mechanical response

Mechanical material + Pneumatic actuation + Digital interaction / V03, V01

A scene to make the idea concrete

A sleeve or costume panel changes its resistance to bending during an interactive performance or simulated task. Flexible motion and deliberate mechanical cues alternate under measured pressure and pose. The proposed garment expresses a digital environment through a calibrated physical response, with user release and comfort constraints built into the interaction rather than inferred from animation.



Operating mechanism

Caltech researchers demonstrated interlocking structured fabrics whose bending stiffness changes when boundary pressure jams their particles [A15R1]. The proposed garment packages small replaceable panels with pressure sensing, pose measurement and controlled venting. Commands select a characterized stiffness range; they do not prescribe arbitrary forces independently of motion and load. A digital scene requests interaction cues through that constrained adapter. Fabric geometry, membrane leakage and loading history alter the response, so each panel needs calibration. The component evidence establishes tunable material behavior, not a completed wearable interaction system.

```

tau_resist = -k(p,h)*(q-q0) - b(p,h)*dq/dt
k(p,h) >= 0; b(p,h) >= 0
P_mech = tau_resist*dq/dt
  
```

This local joint surrogate uses angle q in radians, torque in newton metres and pressure difference p in pascals. History h represents calibrated jamming behavior. Changing stiffness can exchange stored energy, so nonnegative k and b alone do not prove passivity during pressure changes.

Where this framework enters

PM pages 20-21 separate geometric hinges from actuator effort, load and mechanical energy in V03. Pages 16-17 provide V01 scene identity and editable interaction state. UGTS locates panel frames and connections; atomOS records command admission and actual acknowledgments. WANTWOMBAN supplies the feedback architecture, but a pneumatic pressure/flow model explicitly replaces its thermal decoder. The garment's physical pose remains independently observed.

APPLICATION / VALIDATION

A15 / Experiment and future

Clothing with a changing mechanical response

Minimum demonstrator

Start with one panel on an articulated dress form driven through repeatable bending trajectories. Measure torque, angle, pressure, electrical consumption and venting time independently. Characterize flexible, intermediate and stiff settings before connecting a simple digital interaction. Compare requested cues with actual resistance under repeated cycles, small leaks and communication interruptions. Keep garment fit and later user evaluation as separate stages; the initial experiment measures calibrated mechanical interaction on the bench.

Measure	Proposed comparison or criterion
Mechanical cue accuracy	Meet preregistered torque-versus-angle error limits across held-out trajectories, pressures and loading histories, with repeated calibration and uncertainty estimates included.
Release response time	Return to the characterized low-resistance state within the declared interval after release requests, power loss and tested communication failures.
Garment system burden	Measure panel mass, pump energy, acoustic noise and heat during matched tasks, comparing the complete assembly with fixed-stiffness alternatives.

The failure that matters

A leaking membrane can weaken a cue; trapped pressure or jammed particles can delay release. Pose error may place resistance at an unintended point in motion. The implementation therefore requires measured release behavior and a directly accessible venting mechanism; a digital scene reset cannot establish that the fabric has relaxed.

How the future extension would differ

Multiple panels could form costumes, haptic sleeves or adjustable equipment straps whose response follows an editable scene. Scaling would require load-sharing models, washable packaging, replaceable membranes and reliable release under realistic wear. The decisive question is whether a useful range of mechanical cues survives those practical constraints with acceptable bulk, noise and energy.

The exact role of SEED

SEED reproduces digital cue sequences and bench motion schedules. Pressure, fabric history, fit and measured torque remain physical inputs. An identical interaction seed therefore reproduces requested cues only; actual resistance requires its own calibrated comparison and trace.

Component evidence: [A15R1]. See the bibliography for the result supported and the boundary of that evidence.

APPLICATION / DESIGN

A16 / Planetary tools making task fixtures

Planetary equipment that manufactures its own fixtures

Physical fabrication + Robotic assembly + Digital geometry / V05, V07, V01, V06

A scene to make the idea concrete

A planetary work station encounters a sample or instrument that its standard gripper cannot hold securely. It scans the contact geometry, proposes a bounded fixture, manufactures it from accounted feedstock and checks its fit before use. The proposal concerns replaceable task accessories, with finite material and power reserves, rather than a machine reproducing every part of itself.



Operating mechanism

NASA demonstrated printing an uplinked ratchet-wrench design aboard the space station, with the specimen reserved for testing [A16R1]. NASA's ARMADAS separately demonstrated autonomous modular assembly in a terrestrial laboratory [A16R2]. The proposed integration combines measured geometry, constrained fixture generation, fabrication and robotic installation. A contact model determines acceptable shape and load; witness checks determine whether the manufactured part meets that contract. Ground demonstrations and orbital printing support components, while planetary dust, thermal cycling, gravity and feedstock quality remain unresolved environmental qualifications.

```

m_next = m_stock - m_fixture - m_waste
E_next = E_store + E_in - E_fabricate - E_other
task_load <= calibrated_fixture_limit
  
```

Mass terms use kilograms and energy terms joules over one production interval. The load limit is obtained from a declared material, process and test method, not topology alone. Geometry fit, gripping stability and environmental endurance require separate checks before the fixture receives an accepted-use status.

Where this framework enters

PM pages 24-25 provide V05 deposition history, inspection and support admission. Pages 28-29 add V07 command acknowledgment and recovery from partial action. V01 on pages 16-17 supplies bounded editable geometry, while V06 on pages 26-27 accounts for mission resources. UGTS binds the fixture's lineage and interfaces; atomOS distinguishes proposed geometry from inspected hardware. WANTWOMBAN contributes measured process feedback through a manufacturing adapter.

APPLICATION / VALIDATION

A16 / Experiment and future

Planetary equipment that manufactures its own fixtures

Minimum demonstrator

Build a terrestrial bench with a scanner, one polymer printer and a robot that handles instrument surrogates. Predeclare a small family of gripping fixtures and loads. Generate variants from measured interface geometry, print witness specimens and verify dimensional fit and mechanical response independently. Introduce bounded print defects and interrupted installation acknowledgments. Compare fixture production with a stocked set of conventional adapters, accounting for consumed material, energy, time and failed parts.

Measure	Proposed comparison or criterion
Verified task completion	Complete the declared handling tasks with independently checked fit and load limits, recording rejected fixtures and every manual intervention required.
Mission resource accounting	Reconcile stock, finished parts, waste and electrical consumption within measurement uncertainty, including inspection, rejected builds and interrupted production attempts.
Recovery after interruption	Resume from measured hardware state after tested print or installation interruptions, without duplicate commands or silently accepting partially completed fixtures.

The failure that matters

A geometrically accurate fixture can still contain weak bonds, residual stress or contamination. A robot may also lose track of whether installation completed. Recovery must retain partially manufactured and attached objects in the physical scene, and require renewed inspection when material evidence or command acknowledgment is incomplete.

How the future extension would differ

A later station could recycle selected accessories and manufacture supports for cameras, inspection heads or maintenance tasks. Any local-resource feedstock would introduce extraction, sorting, processing and contamination contracts with their own energy costs. The decisive gap is reliable fabrication and verification under the actual planetary environment, especially when replacement sensors and actuators remain externally supplied.

The exact role of SEED

SEED makes candidate geometry searches and simulated placement trials reproducible. Actual scans, feedstock batches, fabrication measurements and installed states remain external records. Repeated seeds select the same proposed fixture only when the complete descriptor and inputs also match.

Component evidence: [A16R1], [A16R2]. See the bibliography for the result supported and the boundary of that evidence.

APPLICATION / DESIGN

A17 / Machines testing candidate designs

Machines that test their own design alternatives

Digital experiment design + Physical testing + Feedback fabrication / V01, V05, V07

A scene to make the idea concrete

A manufacturing cell investigates several replaceable gripper pads or structural inserts before selecting one for a defined task. It generates permitted designs, makes specimens, measures performance and updates its next experiment. The proposed autonomy remains inside a bounded design family and resource budget, with the objective, exclusions and acceptance procedure fixed before the search begins.



Operating mechanism

Boston University's BEAR combined automated fabrication and mechanical experiments with Bayesian selection of subsequent designs [A17R1]. The proposed PM extension makes each candidate a typed, versioned object and connects test evidence to an explicit deployment guard. A predictive model ranks candidates using measured outcomes and uncertainty; fabrication and testing supply new evidence. Simulation can screen designs but cannot overwrite contradictory measurements. The search optimizes a specified task metric within its admitted family, rather than proving that its winner is globally optimal or suitable for untested uses.

```

a_next = argmax_a acquisition(a | measured_data)
a in allowed_designs; experiment_cost(a) <= budget
score(a) = measured_task_value(a) - stated_penalties
  
```

The acquisition rule is a declared algorithm balancing predicted value and uncertainty. Scores and penalties must have compatible units or explicit normalization. Cost may be a constrained resource vector rather than a scalar. These definitions specify experiment selection; they do not establish model accuracy or optimality.

Where this framework enters

PM pages 16-17 give V01 bounded grammar expansion, editable candidate identity and replay. Pages 24-25 give V05 fabrication history and independent inspection. V07 on pages 28-29 tracks asynchronous tests and interrupted operations. UGTS preserves parentage, material and interface contracts; atomOS records candidate admission and committed outcomes. WANTWOMBAN contributes the observed feedback cycle through explicit force, displacement and process-measurement adapters.

APPLICATION / VALIDATION

A17 / Experiment and future

Machines that test their own design alternatives

Minimum demonstrator

Choose one printable pad geometry with two adjustable dimensions and one material. Automate fabrication, specimen identification and a repeatable compression or gripping test using independently calibrated instruments. Predeclare the search budget and evaluation rule. Compare adaptive selection with a grid and seeded random search under equal total resources. Retest finalists on held-out loads and fresh material batches, including repeat specimens that estimate manufacturing variability before any deployment decision.

Measure	Proposed comparison or criterion
Search efficiency	Reach the preregistered held-out performance target using fewer total resources than baseline searches, counting failed fabrication, calibration and repeated specimens.
Independent finalist performance	Confirm selected designs on fresh batches and unseen loads using separate evaluation data, with uncertainty and unsuccessful replications explicitly reported.
Evidence lineage completeness	Trace every accepted score to its exact design, material, instrument calibration, raw measurement and comparison rule, including exclusions and interruptions.

The failure that matters

The search can exploit an instrument artefact, an easy test condition or a flaw in the scoring function. Manufacturing drift can make older evidence misleading. Independent finalist evaluation, calibrated reference specimens and fixed exclusions are therefore part of the experiment contract; an improving training score alone cannot admit a deployed design.

How the future extension would differ

Several cells could share compatible evidence and choose complementary experiments while preserving local material and instrument differences. Transfer between machines would require a measured relation between their processes, rather than merging scores with identical labels. The decisive research question is whether the added contract and lineage machinery improves reproducibility and useful discoveries at the total system level.

The exact role of SEED

SEED records initial candidate sets, randomized baselines and tie-break choices. Physical measurements and failed builds remain immutable observations. Adaptive decisions replay only with those same observations, algorithm versions and ordering; the seed cannot regenerate experimental truth.

Component evidence: [A17R1]. See the bibliography for the result supported and the boundary of that evidence.

APPLICATION / DESIGN

A18 / Wave processing with retained material state

Combined wave-and-material computers

Photonic propagation + Material memory + Digital calibration / V04, V02

A scene to make the idea concrete

An instrument processes optical signals through a programmable wave network whose settings are retained in material cells. Fast propagation performs a selected transformation; slower material updates change its response between tasks. The proposed PM system joins these timescales in one recorded configuration, aiming to reduce repeated conversion or reprogramming for a tightly specified sensing or computation workload.



Operating mechanism

An integrated photonic tensor core has demonstrated convolutional processing using phase-change memory arrays and optical frequency combs [A18R1]. This is direct component evidence for combining waves and retained material states. The proposed extension adds PM's typed routing, explicit calibration and task-dependent update contracts. A physical cell sets a measured optical response; the digital controller writes, verifies and selects configurations. Fast reads and slower writes have different energy and disturbance budgets. Optical power readout and coherent amplitude readout remain distinct, so the chosen architecture must specify which quantities it computes.

```

b(omega) = S(omega; g,T)*a(omega)
dg/dt = F(g,u_write,T); y = CoherentReadout(b;reference)
S^dagger*S <= I [passive, fixed configuration]
  
```

Here a and b are complex power-normalized amplitudes, g material state and T temperature. This proposed baseline measures phase and amplitude against a calibrated coherent reference. Passivity applies at fixed configuration with appropriate ports. Writing or time modulation requires additional accounting; intensity-only detection does not recover this output.

Where this framework enters

PM pages 22-23 give V04 power-normalized wave ports, transfer calibration and passive scaling of the pinion. Pages 18-19 give V02 material history, preparation and measured readout. UGTS connects typed optical and programming ports; atomOS records accepted configurations and calibration identities. WANTWOMBAN supplies an optional four-channel target with explicit scaling and coherent readout; volume preservation does not imply optical gain.

APPLICATION / VALIDATION

A18 / Experiment and future

Combined wave-and-material computers

Minimum demonstrator

Use a small programmable optical matrix with independently measured input/output channels, a stable phase reference, coherent receivers and verified material preparation. Program a limited set of transformations, probe basis inputs and held-out combinations, then repeat after idle periods and temperature changes. Measure write, read, conversion and control energy separately. Compare with a conventional electronic implementation at matched task error, including calibration downtime and failed programming attempts rather than reporting optical propagation time alone.

Measure	Proposed comparison or criterion
Task-level numerical agreement	Meet a preregistered transformation-error limit on held-out inputs across the stated bandwidth and temperature range, preserving explicit readout and scaling conventions.
Retention and disturbance	Bound transfer drift after specified idle periods and read counts, measuring write variability, recalibration frequency and irreversible device changes separately.
Complete computation efficiency	Compare energy and latency at matched accuracy against electronics, including sources, detectors, converters, programming, cooling and digital orchestration for every task.

The failure that matters

Material drift can change the transfer matrix while a digital configuration remains unchanged. Read pulses may disturb memory, and loss or detector saturation can conceal errors. Recalibration must therefore use independent optical observations and invalidate stale settings. A correct seed or configuration hash cannot certify the current physical transform.

How the future extension would differ

Coupling retained weights with volatile nonlinear cells could produce wave-and-material temporal processors that adapt between tasks. Their useful memory, numerical stability and programming endurance would require direct measurement. The decisive question is whether combining propagation and material history improves a complete workload after conversion, calibration and thermal management costs are included, rather than merely moving arithmetic between physical domains.

The exact role of SEED

SEED fixes benchmark inputs, initialization targets and experimental ordering. Actual material states, optical phases, noise and drift require measurements. Exact replay applies to digital commands and recorded inputs; physical transformation agreement uses an explicitly calibrated numerical tolerance.

Component evidence: [A18R1]. See the bibliography for the result supported and the boundary of that evidence.

05 / COMPOSITION

One machine assembled from five ideas

Proposed demonstrator: an adaptive repair workcell. A seeded digital description requests a cradle for a part. A03 reshapes a table; A02 reads history-dependent contact signals; A09 deposits a temporary fixture; A08 schedules the experiment; A04 may later provide a calibrated optical signal path.

Connection	Information crossing the boundary	What must be established
Design -> surface	Target pose, load envelope, frame and tolerances	Reachability and measured load support
Surface -> memory	Timestamped sensor sequence with calibration	Scaling, read disturbance and history preparation
Memory -> decision	Estimated contact/slip state plus uncertainty	A validated decision policy and latency limit
Decision -> fabrication	Bounded path, material ID and support requirements	Actual deposition and independent verification
All -> next design	Measurements, failures and actual resource use	Versioned inputs; no replacement of evidence by predictions

```
total cycle time = sense + decide + move + settle + verify
net useful output = accepted result within the task envelope
```

A useful trial compares the complete workcell against conventional fixtures and ordinary feedback on the same part family. Include setup, calibration, rejects, removal and maintenance. A faster inner computation is only one part of the result.

The demonstration is a research proposal. It is intentionally smaller than a self-maintaining habitat and can fail in ways that reveal which larger integration is premature.

06 / DEVELOPMENT PATH

A sequence of engineering gates

Gate	Evidence required to proceed	Stop or revise when
G0 / specify	Task, units, state, ports, source profile and baseline are explicit	Names conceal missing mechanisms or undefined observables
G1 / characterize	Component response, drift, noise and operating bounds are measured	The selected model cannot explain held-out responses
G2 / couple	Adjacent components exchange correctly typed signals under load	Delay, crosstalk or conversion cost defeats the intended task
G3 / close the loop	The physical loop works over a declared envelope and fault set	Unmodeled events destabilize it or invalidate the guards
G4 / compare	A full-system result exceeds a preregistered baseline or meets a new capability	The apparent advantage disappears after overhead and failures
G5 / extend	Scale and mission claims are supported by new evidence	A small demonstration is being treated as proof of general deployment

These gates specify evidence, not dates. The document does not assign universal development times or technology-readiness scores. Costs, personnel, fabrication access and qualification requirements depend on the selected application and region.

A failed gate is useful information. Preserve the observation and update the model or design. Change an acceptance rule prospectively with a new version; do not relabel a failed run as a pass by altering its original criterion.

Useful first tests isolate the uncertain physical link. For example, measure whether deformation changes a memory readout before building a full robotic skin around it.

Experiments that can reject the idea

Measure the capability the user receives

For a fixture, measure accepted load-bearing geometry and setup time. For a reservoir, measure held-out temporal-task error and total energy. For a wave processor, measure complex transfer accuracy over bandwidth and temperature. For repair, measure the repaired property with an independent method.

Control	Purpose
Conventional baseline	Shows whether ordinary geometry, sensing and control already solve the task
Component removal	Tests whether material memory, source geometry or wave processing adds useful information
Matched resource budget	Prevents extra sensors, training data, energy or manual calibration from being hidden advantages
Held-out conditions	Tests new inputs and environments rather than memorized calibration trajectories
Fault injection	Tests stale measurements, partial actions, device failure and interrupted communication
Independent readout	Checks the actual result with an instrument or method not used to approve it

Report trial count, rejected and missing runs, uncertainty, data partitions, parameter changes and the scope of any averaging. A useful success on one task does not establish universal superiority. Keep computational costs and physical resource costs separate until they are converted into a common, justified objective.

The criteria in A01-A18 are proposed experiment designs. New application-level measurements are not reported in this companion.

Energy, material and irreversible action

$$\Delta E = \int (P_{in} - P_{useful} - P_{loss}) dt$$
$$\Delta M = \int (mass_{in} - mass_{out}) dt$$

recycling: internal transfers cancel in the total boundary

State the boundary before writing a balance. Recovered water enters a usable-water reservoir from a waste/process reservoir; it is not new mass for the whole habitat. Thermal memory consumes or redistributes physical energy. Wave tuning and readout may require significant support electronics.

A physical state has a history

Applied heat remains after a software rejection. Deposited material remains after a failed inspection. An interrupted motion can end between its intended start and finish. Recovery must begin with the actual state, retaining uncertainty until it is resolved.

Application mechanism	Resources and effects to include
Material computation	Driver energy, preparation/reset, readout, drift compensation and cooling
Photonic/phononic processing	Source, tuners, transducers, loss, detection, conversion and control
Adaptive mechanics	Mechanical work, elastic storage, damping, holding and service life
Fabrication/repair	Feedstock, purge, waste, curing, inspection and rework
Distributed operation	Communication, local computation, travel, standby and recovery

The name Perpetuum Mobile describes continuing causal operation in this architecture. Useful physical operation still needs a supplied and accounted resource flow. A recurring digital state does not establish unlimited physical endurance.

PM 1.0 pp. 14, 26-27 and 30 specify the corresponding continuation and interface conditions.

SEED, state and experimental memory

digital replay = seed + immutable descriptor + initial state
+ ordered input transcript + numeric rules
physical repeatability = prepared state + calibration
+ measured disturbances + uncertainty

A seed can choose stimuli, procedural geometry, task order, optimization initialization and controlled simulation scenarios. Use stable logical identities for its streams; do not derive them from changing worker IDs or network arrival order.

The complete PPM-SEED-v1 construction is retained in the embedded PM 1.0 package. It fixes byte framing, SHA-256 domains, counters, range sampling and transcripts. Its fixture results validate that reference construction on the recorded environments; they do not validate the new machines in this book.

A minimum useful experiment record

Record	Contents
Configuration	Application ID, source/profile hashes, geometry/material/firmware versions
Initial state	Digital state and physical preparation with measured tolerance
Inputs	Seed-derived draws and actual external measurements with timestamps
Actions	Proposals, guards, actual commands, acknowledgments and partial actions
Results	Raw readouts, uncertainty, outcome and independent comparison
Receipt	Terminal event count, artifact hashes, code/environment and missing evidence

This companion adds content, source and rendering verification. Existing seed/calculation receipts remain source evidence with their original scope. No new claim of physical validation follows from the PDF passing editorial or integrity checks.

Hash consistency supports comparison with a known expected artifact. Unsigned hashes do not authenticate authorship or establish that a measurement is true.

10 / SYNTHESIS

A research portfolio with dependencies

Research cluster	Applications	Shared bottleneck
Embodied surfaces	A02, A03, A06, A11, A15	Calibration under changing load, contact, temperature and shape
Wave/material computation	A02, A04, A07, A18	Useful signal processing after conversion, drift and control overhead
Fabrication and discovery	A05, A08, A09, A16, A17	Reliable measurement of what was actually made and what property it has
Persistent distributed operation	A01, A10, A12, A13, A14	Maintaining trustworthy state under delay, failure, degradation and resource limits

The main research opportunity is composition. A geometry system can describe a support, a material system can remember a signal, and a controller can act on a measurement. Combining them becomes useful only when the interfaces preserve the meaning, timing and uncertainty required by the task.

The broadest future machine would inspect its surroundings, identify a bounded need, synthesize a change, enact it and verify the outcome. The strongest starting point is an experiment where one uncertain interface can be measured independently.

A surface that remembers physically, reshapes mechanically, processes signals optically and coordinates repair is a compelling direction. The hypotheses in this book turn that direction into a set of experiments capable of showing where it works, where it fails and whether it earns its complexity.

This is a design and research companion. It is not a novelty determination, production certification, procurement recommendation or a report of completed application hardware.

APPENDIX / CONVERSATION WITNESS

Original reply / 1-6

The following is the original assistant application reply preserved as a source witness. Its opening commit receipt refers to ea9941d, the preceding PM 1.0 delivery, not this companion.

These are proposed integrations. My strongest picks for transformative potential are the first three.

1 / Habitats that inspect and repair themselves

Material sensors detect changes, robots investigate, and fabrication systems construct measured repairs. The same model accounts for energy, water, replacement material and waste. The hardest problem is verifying that a repair is structurally sound.

2 / Robot skin that computes through its material

Physical memory retains recent touch, temperature and deformation. A trained readout recognizes slipping or changing loads, then nearby actuators reshape the surface. The skin becomes part of the sensing and computation system.

3 / A workshop table that becomes its own tooling

A programmable surface reshapes into supports, cradles, molds or assembly fixtures. Cameras and load sensors verify the shape; a printer adds features the moving surface cannot produce.

4 / A programmable optical nervous system

Photonic networks route signals and perform calibrated matrix operations. When a component drifts or fails, the system measures the change and searches for another usable configuration.

5 / Remote manufacturing from a “construction seed”

Send a compact design, ruleset and acceptance criteria to a remote workshop. Its machines reconstruct the design, adapt it to measured surroundings and record every physical deviation.

6 / VR environments with physical, moving geometry

A seeded virtual world coordinates with actuated surfaces and props, creating tangible controls, terrain or objects. Reachability, tracking and human-contact limits would constrain each transformation.

APPENDIX / CONVERSATION WITNESS

Original reply / 7-12

7 / Rooms with programmable acoustics

Movable panels and tunable acoustic elements reshape measured sound propagation for different listening zones, performances or sensing tasks.

8 / Laboratories that reconfigure their own experiments

Robotic fixtures position samples, programmable optics change illumination, and fabrication produces custom holders or channels. Seeded stimuli make successive experiments comparable.

9 / Printers that grow structures under continuous inspection

Each branch is deposited, scanned and checked before the next branch depends on it. Failed geometry becomes a correction or rework task with an explicit material history.

10 / Persistent worlds synchronized through rules and edits

Devices regenerate a shared environment from a seed and versioned definitions, then exchange edits and observations attached to stable object identities.

11 / Deployable structures that retune themselves

Instrument mounts, shades or support structures unfold through guarded hinge sequences, then adjust to measured temperature, load and vibration.

12 / Repair swarms that tolerate communication gaps

Robots share reproducible plans but maintain local physical control. When communication returns, they reconcile observations, interrupted actions and completed repairs.

APPENDIX / CONVERSATION WITNESS

Original reply / 13-18

13 / Infrastructure with distributed physical memory

Sensor networks retain patterns of loading and thermal exposure. Those histories help select where bridges, machines or structures should receive closer inspection.

14 / Greenhouses that coordinate survival and maintenance

Irrigation, heating, storage and recovery share a resource model. Robots inspect leaks or blocked channels and manufacture bounded replacements when appropriate.

15 / Clothing with a changing mechanical response

Garments could adjust shape, support or thermal behavior using local sensing and actuation. Material memory could help distinguish brief contact from sustained loading.

16 / Planetary equipment that manufactures its own fixtures

A rover measures an unfamiliar surface, generates a suitable support or tool attachment, fabricates it and verifies its fit before use.

17 / Machines that test their own design alternatives

A digital model proposes a bounded modification; a fabrication system builds it; instruments measure the result; the next design incorporates that evidence.

18 / Combined wave-and-material computers

Fast optical or acoustic mixing feeds slower material memory, with digital control handling configuration and readout. The central research question is whether the complete system gains anything after conversion, calibration and control costs.

The most distinctive combination: a surface that can remember physically, reshape mechanically, process signals optically and coordinate its own repair.

Your framework could provide the common language connecting those capabilities. Improvements in efficiency, reliability or autonomy would still need comparison with conventional implementations.

REFERENCES

Component research / 1

Primary component sources checked 15 September 2026. Chapter equations and complete PM integrations are proposals unless a specific result is attributed. A component paper does not validate an entire application.

A01R1

NASA: Integrated System for Autonomous and Adaptive Caretaking [Source](#).

Supports: Robotic inspection combined with vehicle telemetry and coordinated execution.

Boundary: The cited ISS obstruction scenario did not include manipulator repair.

A01R2 / A05R2 / A09R1

Buchner et al.: Vision-controlled jetting for composite systems and robots [Source](#).

Supports: Measured print geometry informs subsequent deposition.

Boundary: This is manufacturing evidence, not autonomous habitat repair validation.

A02R1

Tan et al.: Tactile sensory coding and learning with bio-inspired optoelectronic spiking afferent nerves [Source](#).

Supports: Pressure sensing, optical spike encoding and photomemristor integration.

Boundary: No validation of this proposed deforming robot skin or grip controller.

A02R2 / A13R2

Du et al.: Reservoir computing using dynamic memristors for temporal information processing [Source](#).

Supports: Experimental material dynamics used for temporal processing.

Boundary: Software readout and specific devices; not a universal material law.

A03R1

Follmer et al.: inFORM - Dynamic Physical Affordances and Constraints through Shape and Object Actuation [Source](#).

Supports: Shape displays can provide changing constraints and actuate objects.

Boundary: The work does not qualify load-bearing tooling or deposited fixtures.

REFERENCES

Component research / 2

Primary component sources checked 15 September 2026. Chapter equations and complete PM integrations are proposals unless a specific result is attributed. A component paper does not validate an entire application.

A04R1

Perez et al.: Multipurpose silicon photonics signal processor core [Source](#).

Supports: Experimental programmable waveguide routing and linear transformations.

Boundary: No validation of the PM compiler or its pinion mapping.

A04R2

Perez-Lopez et al.: Multipurpose self-configuration of programmable photonic circuits [Source](#).

Supports: Measured feedback used for characterization, routing and configuration.

Boundary: Component experiments do not guarantee convergence for arbitrary requested transforms.

A05R1 / A16R2

NASA: Robot Team Builds High-Performance Digital Structure [Source](#).

Supports: Ground demonstration of coordinated autonomous structural block assembly.

Boundary: This does not establish deployed space construction or construction-seed semantics.

A06R1

Suzuki et al.: RoomShift - Room-scale Dynamic Haptics for VR with Furniture-moving Swarm Robots [Source](#).

Supports: Demonstrated furniture-moving robots as physical VR proxies.

Boundary: Not a guarantee of arbitrary room layouts or unrestricted human contact.

A06R2

HapticBots: Distributed Encountered-type Haptics for VR with Multiple Shape-changing Mobile Robots [Source](#).

Supports: Research prototype with moving and shape-changing haptic proxies.

Boundary: Does not validate PM mapping, readiness or cross-room replay contracts.

REFERENCES

Component research / 3

Primary component sources checked 15 September 2026. Chapter equations and complete PM integrations are proposals unless a specific result is attributed. A component paper does not validate an entire application.

A07R1

Optimizing multi-user indoor sound communications with acoustic reconfigurable metasurfaces (2024) [Source](#).

Supports: Demonstrated room-channel shaping with tunable resonators.

Boundary: Does not validate this adaptive occupied-room architecture.

A08R1

Burger et al., A mobile robotic chemist, Nature (2020) [Source](#).

Supports: Autonomous operation across defined laboratory workstations.

Boundary: Does not establish arbitrary laboratory reconfiguration.

A10R1

OpenUSD 25.11: Sdf Scene Description Foundations [Source](#).

Supports: Layered scene description and explicit composition.

Boundary: Does not itself guarantee network convergence or intent preservation.

A11R1

Chen and Shea, An Autonomous Programmable Actuator and Shape Reconfigurable Structures Using Bistability and Shape Memory Polymers [Source](#).

Supports: Programmed thermal deployment and reconfigurable structures.

Boundary: Does not establish this repeated feedback-trimming architecture.

A12R1

NASA Jet Propulsion Laboratory: NeBula [Source](#).

Supports: Multi-robot autonomy with intermittent mesh communication.

Boundary: Does not establish autonomous repair under arbitrary partitions.

REFERENCES

Component research / 4

Primary component sources checked 15 September 2026. Chapter equations and complete PM integrations are proposals unless a specific result is attributed. A component paper does not validate an entire application.

A13R1

Cambridge: Staffordshire Bridges long-term monitoring [Source](#).

Supports: Installed fibre sensors, continuous acquisition and digital-twin analysis on real bridges.

Boundary: Does not demonstrate distributed material-memory modules or autonomous structural diagnosis.

A14R1

WUR: Digital Twin controls cucumber cultivation [Source](#).

Supports: A reported cultivation trial links models and sensors to climate, irrigation and staff pruning instructions.

Boundary: Does not demonstrate the proposed equipment-repair scheduler or autonomous handling of every biological disturbance.

A15R1

Wang et al.: Structured fabrics with tunable mechanical properties [Source](#).

Supports: Experiments demonstrate reversible pressure-controlled stiffening of interlocking structured fabrics.

Boundary: Does not qualify a finished garment, medical application, impact protection or unrestricted user contact.

A16R1

NASA: Space-station printer builds a ratchet wrench [Source](#).

Supports: An uplinked design was manufactured in orbit during a technology demonstration.

Boundary: The reported wrench was reserved for ground testing; this was not autonomous planetary fixture qualification.

A17R1

Gongora et al.: A Bayesian experimental autonomous researcher [Source](#).

Supports: Automated experiments and Bayesian optimization explored a defined mechanical design family.

Boundary: Does not establish unrestricted machine redesign, global optimality or qualification for arbitrary deployed tasks.

A18R1

Feldmann et al.: Integrated photonic tensor core [Source](#).

Supports: Demonstrates photonic in-memory convolution using phase-change arrays and optical frequency combs.

Boundary: Does not establish the proposed PM controller, arbitrary transforms or universal total-system efficiency advantages.

PROVENANCE

Provenance and editable package

Framework attribution: Tom Klootwijk / NL200678942.

Edition: Frontier Applications A1.0, 15 September 2026.

Preparation: Codex-assisted expansion of the preceding assistant reply, using the supplied framework and primary component research.

Normative base specification

Perpetuum Mobile PM 1.0 / 38 pages. The exact original is embedded as a source witness, including its nested four-source package. All PM page references in this companion refer to that edition.

```
b6532f9638e6cca77d75c2636a54afb5  
daf587a4e69aceff72102384252a5a55
```

Original reply witness

original_reply.md preserves the prior reply including its historical commit link. Its exact SHA-256 is:

```
48de58ee743d06fb7234ef19fa5bcfb5  
ae1e5a1a4e563bc1b436e2cd05eaba40
```

Package contents

Embedded files include this book's editable Markdown and JSON, all eighteen structured application chapters, bibliography, builder, original reply, source/attachment manifests, cover image and exact imagegen prompt. The base PM 1.0 PDF is embedded unchanged.

The cover was generated with the built-in image_gen tool; its asset and prompt are retained in the repository. Native vector process diagrams are explanatory layouts. Integrity records are unsigned and are not claims of independent authorship verification or scientific validity.